20261026 Quiz

for (MATH210) Applied Complex Variables

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Problem 1. Find all $z \in \mathbb{C}$ such that $\sin z = 1$.

Solution. Using the exponential definition of the complex sine, set $w = e^{iz} \neq 0$. Then

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 1 \quad \Longleftrightarrow \quad w^2 - 2iw - 1 = (w - i)^2 = 0.$$

Thus $e^{iz} = i$. Writing $z = x + iy$, we have

$$e^{iz} = e^{-y}(\cos x + i \sin x) = i.$$

Comparison of moduli gives $e^{-y} = 1$, hence $y = 0$; comparison of arguments then gives $x = \pi/2 + 2\pi k$ for $k \in \mathbb{Z}$. Therefore all solutions are

$$\boxed{z = \frac{\pi}{2} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

□

Problem 2. Find $a, b \in \mathbb{R}$ such that

$$\frac{(\sqrt{3} + i)^{20}}{(1 - i)^{10}} = a + bi.$$

Solution. The polar forms of the two bases are

$$\sqrt{3} + i = 2e^{i\pi/6}, \qquad 1 - i = \sqrt{2}\, e^{-i\pi/4}.$$

Consequently, by De Moivre's theorem,

$$\begin{aligned} \frac{(\sqrt{3} + i)^{20}}{(1 - i)^{10}} &= \frac{2^{20} e^{i10\pi/3}}{2^5 e^{-i5\pi/2}} = 2^{15} e^{i35\pi/6} \\ &= 2^{15} \left(\frac{\sqrt{3}}{2} - \frac{i}{2} \right) = 2^{14}(\sqrt{3} - i). \end{aligned}$$

Hence

$$\boxed{a = 2^{14}\sqrt{3}, \qquad b = -2^{14}.}$$

□

Problem 3. Find an entire function $f(z)$ such that

$$\operatorname{Re} f(z) = x^2 - y^2 + y, \qquad z = x + iy,$$

and $f(0) = i$.

Solution 1. Write $f = u + iv$, where

$$u(x, y) = x^2 - y^2 + y.$$

By the Cauchy–Riemann equations,

$$v_x = -u_y = 2y - 1, \qquad v_y = u_x = 2x.$$

Integrating the first equation with respect to x, we obtain

$$v(x, y) = (2y - 1)x + \phi(y).$$

Differentiating with respect to y and using $v_y = 2x$ gives

$$2x + \phi'(y) = 2x,$$

so $\phi(y) = c$ for some real constant c. Consequently,

$$f(z) = (x^2 - y^2 + y) + i(2xy - x + c) = z^2 - iz + ic.$$

Since $f(0) = i$, we have $c = 1$. Thus

$$\boxed{f(z) = z^2 - iz + i.}$$

This polynomial is entire, and its real part and value at 0 satisfy the given conditions.

□

Solution 2. Observe that

$$\operatorname{Re}(z^2) = x^2 - y^2, \qquad \operatorname{Re}(-iz) = y.$$

Hence $z^2 - iz$ has exactly the prescribed real part. Adding a purely imaginary constant does not change that real part, and the condition $f(0) = i$ fixes the constant. Therefore

$$\boxed{f(z) = z^2 - iz + i.}$$

Being a polynomial, f is entire, and direct substitution verifies both requirements.

□

Problem 4. Evaluate

$$\lim_{R \to \infty} \int_{C_R} \frac{2z^2 - 1}{z^4 + z^2 + 1} \, \mathrm{d}z,$$

where C_R is the circle $|z - 1| = R$ with counterclockwise orientation.

Solution. For $z \in C_R$, the reverse triangle inequality gives

$$R - 1 \le |z| \le R + 1.$$

For sufficiently large R, in particular for $R \ge 5$, the denominator satisfies

$$\begin{aligned} |z^4 + z^2 + 1| &\ge |z|^4 - |z|^2 - 1 \\ &\ge (R - 1)^4 - (R + 1)^2 - 1 > 0, \end{aligned}$$

while $|2z^2 - 1| \le 2(R+1)^2 + 1$. Since the length of C_R is $2\pi R$, the ML inequality yields

$$\left| \int_{C_R} \frac{2z^2 - 1}{z^4 + z^2 + 1} \, \mathrm{d}z \right| \le \frac{2\pi R(2(R+1)^2 + 1)}{(R-1)^4 - (R+1)^2 - 1} \longrightarrow 0.$$

Therefore

$$\boxed{\lim_{R \to \infty} \int_{C_R} \frac{2z^2 - 1}{z^4 + z^2 + 1} \, \mathrm{d}z = 0.}$$

□

Problem 5.

(a) Evaluate

$$\int_C \frac{e^{z^2}}{z} \, \mathrm{d}z,$$

where C is the circle $|z| = 1$, oriented counterclockwise.

(b) Use part (a) to evaluate

$$\int_0^{2\pi} e^{\cos 2\theta} \cos(\sin 2\theta) \, \mathrm{d}\theta.$$

Solution.

(a) The function $g(z) = e^{z^2}$ is entire, and C is positively oriented with 0 in its interior. By the Cauchy integral formula,

$$\boxed{\int_C \frac{e^{z^2}}{z} \, \mathrm{d}z = 2\pi i \, g(0) = 2\pi i.}$$

(b) Parametrize C by $z = e^{i\theta}$, $0 \le \theta \le 2\pi$, so that $\mathrm{d}z/z = i \, \mathrm{d}\theta$. Part (a) becomes

$$2\pi i = i \int_0^{2\pi} e^{e^{2i\theta}} \, \mathrm{d}\theta.$$

Euler's formula gives the essential decomposition

$$e^{e^{2i\theta}} = e^{\cos 2\theta}\big(\cos(\sin 2\theta) + i \sin(\sin 2\theta)\big).$$

Dividing by i and taking real parts, we obtain

$$\boxed{\int_0^{2\pi} e^{\cos 2\theta} \cos(\sin 2\theta) \, \mathrm{d}\theta = 2\pi.}$$

□

Problem 6. Evaluate

$$\int_C \frac{e^z + \cos z}{(z - i)^{2025}} \, \mathrm{d}z,$$

where C is the boundary of the square with vertices $4\exp(i2k\pi/4)$ $(k = 0, 1, 2, 3)$, oriented clockwise.

Solution. The square has vertices $4, 4i, -4, -4i$, and i lies in its interior. Let

$$g(z) = e^z + \cos z,$$

which is entire. The generalized Cauchy integral formula, with a minus sign because C is oriented clockwise, gives

$$\int_C \frac{g(z)}{(z-i)^{2025}}\,\mathrm{d}z = -\frac{2\pi i}{2024!}g^{(2024)}(i).$$

Since 2024 is divisible by 4, repeated differentiation of the cosine gives

$$g^{(2024)}(z) = e^z + \cos z.$$

Therefore

$$\boxed{\int_C \frac{e^z+\cos z}{(z-i)^{2025}}\,\mathrm{d}z = -\frac{2\pi i}{2024!}(e^i+\cos i).}$$

□

Problem 7. Let $P(z)$ be a nonconstant polynomial. Describe the image of the set

$$A := \{z \in \mathbb{C} \mid P(z) \neq 0\}$$

under the map $f(z) = \exp(1/P(z))$.

Solution. By the fundamental theorem of algebra, every $w \in \mathbb{C}$ is attained by P: indeed, the nonconstant polynomial $P(z) - w$ has a root. Restricting to A therefore gives

$$P(A) = \mathbb{C} \setminus \{0\}, \qquad \left\{\frac{1}{P(z)} \mid z \in A\right\} = \mathbb{C} \setminus \{0\}.$$

The exponential never vanishes, so $f(A) \subseteq \mathbb{C} \setminus \{0\}$. Conversely, take any $w \neq 0$. A logarithm L of w can be chosen nonzero: if $w \neq 1$, every logarithm is nonzero; if $w = 1$, choose $L = 2\pi i$. Since $L \neq 0$, there exists $z \in A$ with $1/P(z) = L$, and then $f(z) = e^L = w$. Hence

$$\boxed{f(A) = \mathbb{C} \setminus \{0\}.}$$

□

Problem 8. Consider the power series

$$\sum_{n=1}^{\infty} a_n z^n, \qquad a_n = \begin{cases} n & \text{if } n \text{ is even,} \\ 1/n & \text{if } n \text{ is odd.} \end{cases}$$

(a) Find the radius of convergence of the series.
Hint. You may evaluate $\lim_{n\to\infty} a_n^{1/n}$.

(b) Find all z at which the series converges.

Solution.

(a) We first note that

$$\ln(n^{1/n}) = \frac{\ln n}{n} \longrightarrow 0, \qquad \text{so} \qquad n^{1/n} \longrightarrow 1.$$

Along the even and odd indices, respectively,

$$a_n^{1/n} = n^{1/n} \longrightarrow 1, \qquad a_n^{1/n} = n^{-1/n} \longrightarrow 1.$$

Hence $\lim_{n\to\infty} a_n^{1/n} = 1$. For each fixed $z \in \mathbb{C}$,

$$\lim_{n\to\infty} |a_n z^n|^{1/n} = |z|.$$

By the root test, the series converges absolutely for $|z| < 1$ and diverges for $|z| > 1$. Thus its radius of convergence is

$$\boxed{R = 1}.$$

(b) For $|z| = 1$, the even-index terms satisfy

$$|a_{2k} z^{2k}| = 2k \not\longrightarrow 0.$$

Thus the terms fail to approach zero, and the series diverges at every point of the unit circle. Combining this with part (a), its convergence set is

$$\boxed{\{z \in \mathbb{C} \mid |z| < 1\}}.$$

□

Problem 9. Show that the series

$$\sum_{n=1}^{\infty} \frac{e^{inz}}{n^2}$$

converges uniformly on $\operatorname{Im} z \geq 0$.

Solution 1. For $z = x + iy$ with $y = \operatorname{Im} z \geq 0$,

$$\left| \frac{e^{inz}}{n^2} \right| = \frac{e^{-ny}}{n^2} \leq \frac{1}{n^2}.$$

Since $\sum_{n=1}^{\infty} 1/n^2$ converges and the bound is independent of z, the Weierstrass M-test proves that the series converges uniformly on $\{z \in \mathbb{C} \mid \operatorname{Im} z \geq 0\}$.

□

Solution 2. Let $D = \{z \in \mathbb{C} \mid \operatorname{Im} z \geq 0\}$ and

$$S_N(z) = \sum_{n=1}^{N} \frac{e^{inz}}{n^2}.$$

For each $z \in D$, the inequality

$$\left| \frac{e^{inz}}{n^2} \right| \leq \frac{1}{n^2}$$

shows that the series converges absolutely; denote its sum by $S(z)$. For $N \geq 1$ and every $z \in D$,

$$\begin{aligned} |S(z) - S_N(z)| &\leq \sum_{n=N+1}^{\infty} \frac{1}{n^2} \\ &\leq \sum_{n=N+1}^{\infty} \frac{1}{n(n-1)} \\ &= \sum_{n=N+1}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \frac{1}{N}. \end{aligned}$$

Given $\varepsilon > 0$, choose an integer $N_0 > 1/\varepsilon$. Then for all $N \geq N_0$ and $z \in D$,

$$|S(z) - S_N(z)| \leq \frac{1}{N} \leq \frac{1}{N_0} < \varepsilon.$$

This proves uniform convergence on D directly from its definition.

□

Problem 10. Determine whether each statement is true or false, and justify your answer.

(a) There exists an analytic function $f(z)$ with $\operatorname{Re} f(z) = x^3 - y^3$, where $z = x + iy$.

(b) If $f(z)$ is entire and $|f(z)| \geq C > 0$ for all $z \in \mathbb{C}$, then f is a constant function.

Solution.

(a) **False.** The real part u of an analytic function is harmonic. But for $u(x, y) = x^3 - y^3$,

$$u_{xx} + u_{yy} = 6x - 6y,$$

which is not identically zero on any nonempty open set. Thus u cannot be the real part of an analytic function on an open domain.

(b) **True.** Since $|f(z)| \geq C > 0$, the entire function f never vanishes. Hence $h(z) = 1/f(z)$ is entire and

$$|h(z)| = \frac{1}{|f(z)|} \leq \frac{1}{C} \qquad (z \in \mathbb{C}).$$

Liouville's theorem makes h constant. Its constant value is nonzero, so $f = 1/h$ is constant as well.

□